Heated cylinder convection

EMAE 285 · Lab 3

Experiment overview

Measured electrical heating input, cylinder and ambient temperatures, and airflow speed to estimate convective heat transfer. Compared Reynolds/Nusselt behavior with an empirical crossflow correlation.

Variables and instrumentation

Voltage, current, surface temperature, ambient temperature, airspeed

Analysis and presentation

Heat-transfer coefficient versus airflow; Nu–Re comparison; uncertainty

Engineering relevance

Evidence of thermal instrumentation and energy-balance reasoning relevant to cooling and equipment engineering.

Scope and attribution

Course laboratory / group work unless noted. The resonance experiment is a proposed design and has no claimed measured outcomes.

Portfolio case-study brief based on coursework and user-provided descriptions. This is NOT a copy of the original laboratory report. Diagrams on the website are explanatory illustrations, not photographs or original experimental plots.

Original source (may require Google Drive permission):
https://docs.google.com/document/d/1ZJYzEeU27WXz68gYJs_AWHPUi6mRfasXHt4_GFkP3x4/edit